Race strategy Monte Carlo

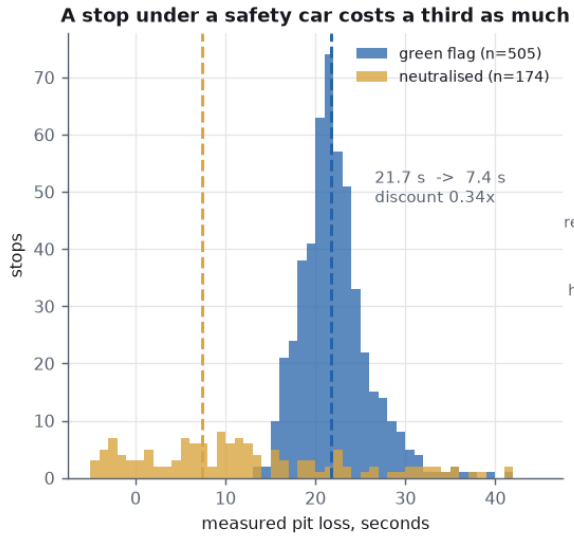
Pit strategy solved from 24,386 F1 lap times and 912 pit stops, 2023 — with every input measured from the same season the answer is then tested against.

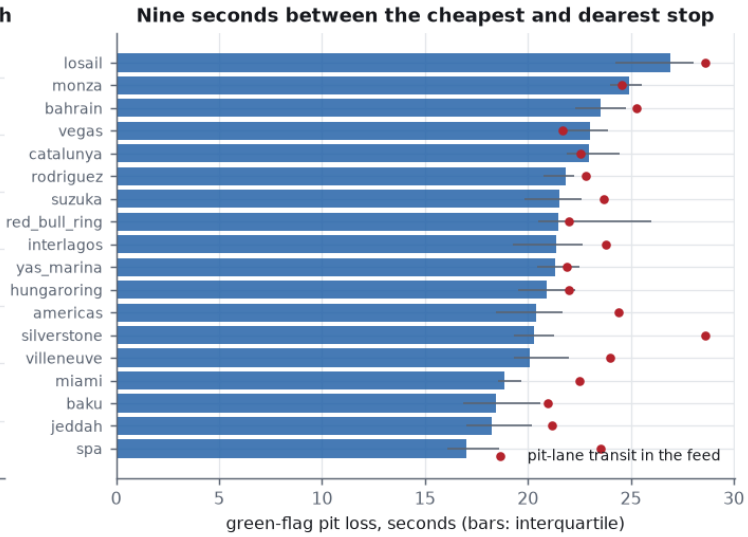
The model recommends too few stops at 9 of 14 circuits

It agrees at 5 and recommends too many at 0. An error with a direction is a missing term, not noise — and the term it is missing is one the lap times cannot measure, because teams stop before it bites.

21.7 s	7.4 s	+0.0316 s/lap	5/14
green-flag pit loss	under a safety car	degradation	circuits the model gets right

Everything the optimiser is fed is measured from the same lap times it is then tested against: the pit loss, the degradation, the safety-car hazard. Nothing is borrowed from a published figure.





What a pit stop costs

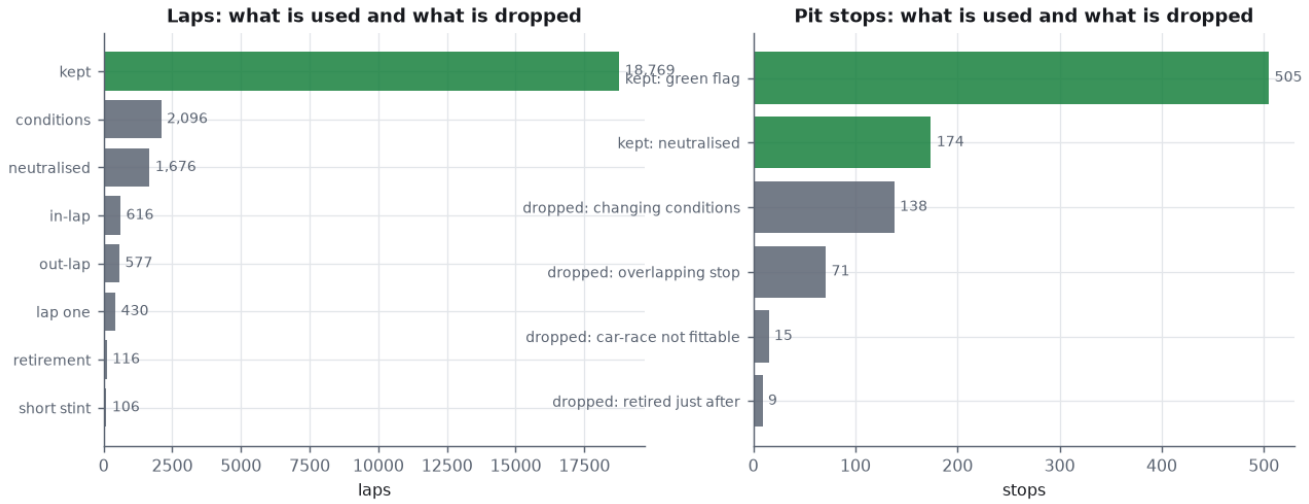
The feed records a pit-lane transit time, not a pit loss.

The cost is measured instead as the excess of the in-lap and the out-lap over what the car's own fitted line says those laps should have taken: 21.75 s under green flag across 505 stops, interquartile 19.75 to 23.84. The same measurement over the 174 stops taken while the race was neutralised gives 7.43 s, a discount of 0.34×. That discount is the only reason a strategy model needs a simulation at all.

circuit	pit loss	interquartile	transit in feed	stops
losail	26.90	24.23 to 28.04	28.59	49
monza	24.92	24.01 to 25.54	24.54	25
bahrain	23.51	22.32 to 24.79	25.27	42
vegas	23.00	21.54 to 23.89	21.69	16
catalunya	22.96	21.89 to 24.45	22.57	43
rodriguez	21.84	20.78 to 22.26	22.82	17
suzuka	21.55	19.82 to 22.62	23.67	28
red_bull_ring	21.49	20.48 to 25.99	21.98	28
interlagos	21.38	19.25 to 22.68	23.81	33
yas_marina	21.32	20.45 to 22.52	21.90	36
hungaroring	20.94	19.54 to 22.29	22.00	36
americas	20.41	18.46 to 21.66	24.41	36
silverstone	20.31	19.34 to 21.28	28.61	12
villeneuve	20.10	19.34 to 22.00	23.98	20
miami	18.88	18.55 to 19.67	22.48	19
baku	18.43	16.84 to 20.61	20.97	11
jeddah	18.26	16.99 to 20.22	21.15	11
spa	17.02	16.11 to 18.60	23.53	37

Which laps and stops are used

Single-car pace scatter is 0.677 s a lap, measured as the residual spread of one car around its own line. The pooled residual from a fit with car-race fixed effects is 1.608 s, which is larger because it still carries differences between cars inside a race; a one-car simulation has no business inheriting those.

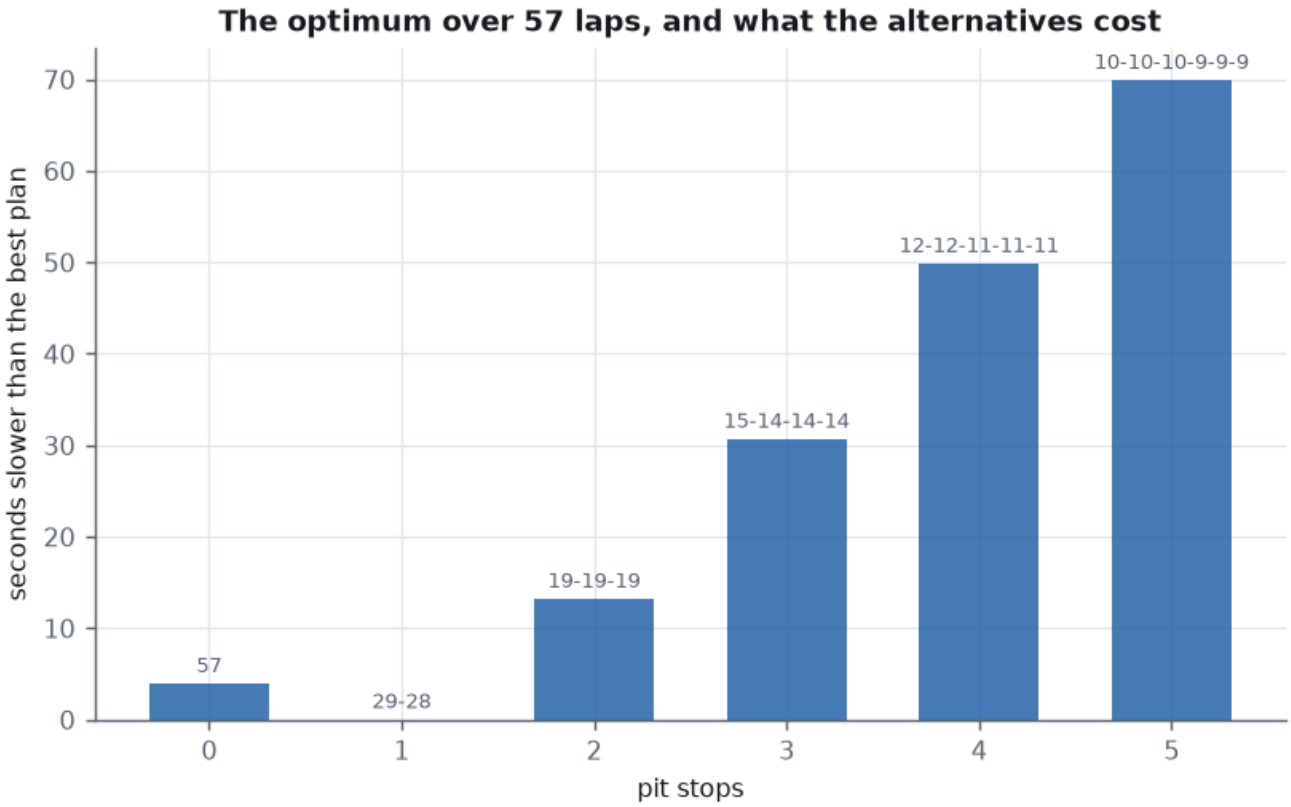


rule	laps	share
lap one	430	1.76%
neutralised	1,676	6.87%
in-lap	616	2.53%
out-lap	577	2.37%
retirement	116	0.48%
conditions	2,096	8.60%
short stint	106	0.43%
kept	18,769	76.97%

The race as arithmetic

Race time splits into a part no plan can change and a part that is entirely the plan: the sum of squared stint lengths, and the stops. Two things follow without simulating anything. Equal stints minimise the sum of squares, so the best plan at any stop count is the balanced split. And differentiating gives the best number of stints outright, k* = N√(d/2L), which at the pooled numbers over 57 laps is 1.537.

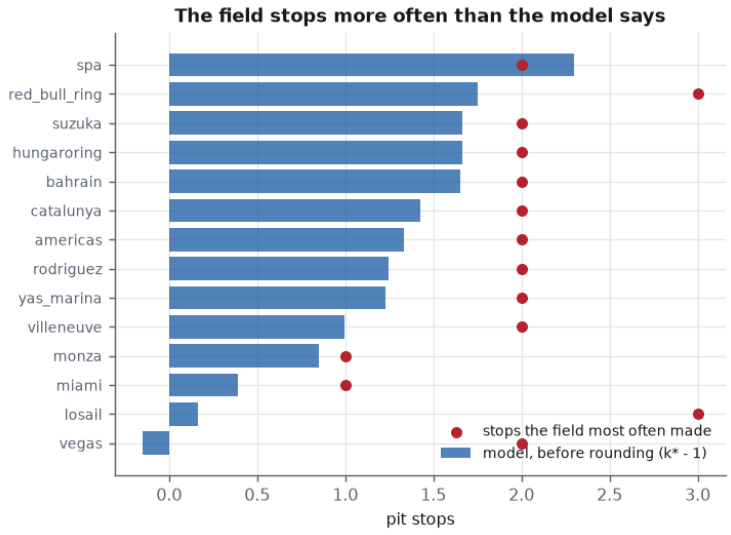
The exhaustive search over every partition of the distance finds exactly what the algebra does, which is a test rather than a hope. Across the degradation confidence interval the optimum moves between 1 and 2 stints, so on the pooled number the data does NOT pick a stop count on its own.



stops	plan	race time, s	behind the best
0	57	5376.97	+3.92
1	29-28	5373.05	+0.00
2	19-19-19	5386.23	+13.18
3	15-14-14-14	5403.71	+30.66
4	12-12-11-11-11	5422.90	+49.84
5	10-10-10-9-9-9	5442.93	+69.88

Tested against what the field actually did

Agrees at 5, recommends too few stops at 9, too many at 0. The direction is the finding. Random error would land on both sides.





circuit	laps	degradation	95% interval	pit loss	k*	model	field	verdict
americas	56	+0.0709	+0.0589 to +0.0830	20.41	2.33	1	2	too few stops
bahrain	57	+0.1020	+0.0710 to +0.1329	23.51	2.65	2	2	agrees
catalunya	66	+0.0620	+0.0505 to +0.0736	22.96	2.43	1	2	too few stops
hungaroring	70	+0.0606	+0.0488 to +0.0723	20.94	2.66	2	2	agrees
losail	57	+0.0225	-0.0012 to +0.0463	26.90	1.17	0	3	too few stops
miami	57	+0.0225	+0.0144 to +0.0307	18.88	1.39	0	1	too few stops
monza	51	+0.0657	+0.0557 to +0.0757	24.92	1.85	1	1	agrees
red_bull_ring	71	+0.0646	+0.0530 to +0.0763	21.49	2.75	2	3	too few stops
rodriguez	71	+0.0436	+0.0346 to +0.0526	21.84	2.24	1	2	too few stops
spa	44	+0.1912	+0.1511 to +0.2312	17.02	3.30	2	2	agrees
suzuka	53	+0.1091	+0.0880 to +0.1303	21.55	2.67	2	2	agrees
vegas	50	+0.0133	-0.0102 to +0.0369	23.00	0.85	0	2	too few stops
villeneuve	70	+0.0328	+0.0154 to +0.0501	20.10	2.00	1	2	too few stops
yas_marina	58	+0.0628	+0.0559 to +0.0697	21.32	2.23	1	2	too few stops

The circuits that are refused rather than ranked

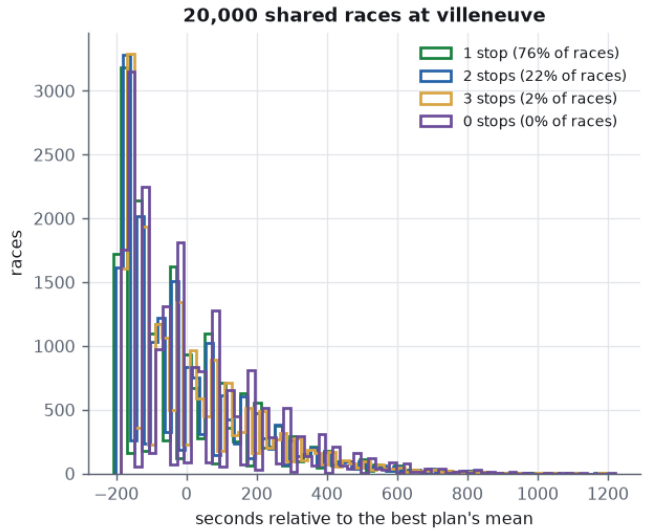
Their measured degradation is not positive — on those laps tyres apparently improve with age, and no stop count exploits that. Returning a strategy anyway would be the optimiser lending false precision to a number that has none.

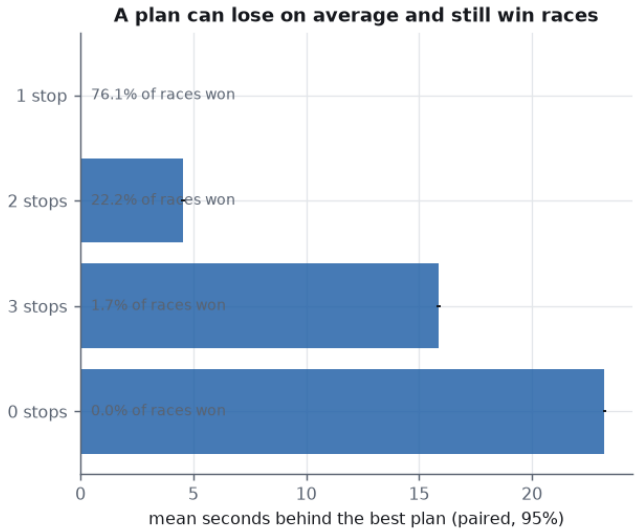
circuit refused	degradation	95% interval	field ran
baku	-0.0047	-0.0506 to +0.0411	1
interlagos	-0.0099	-0.0441 to +0.0244	4
jeddah	-0.0256	-0.0591 to +0.0079	1
silverstone	-0.0032	-0.0260 to +0.0197	1

Is it safety car risk?

20,000 simulated races at villeneuve, every plan run through the same draws so the comparison is paired.

The simulation reproduces the observed frequency of neutralisation without being told it: 75.4% of simulated races contain at least one, against 65.0% of the real dry races. Reacting to a neutralisation — bringing the next stop forward when one starts within the window — is worth +1.24 s to the best plan.

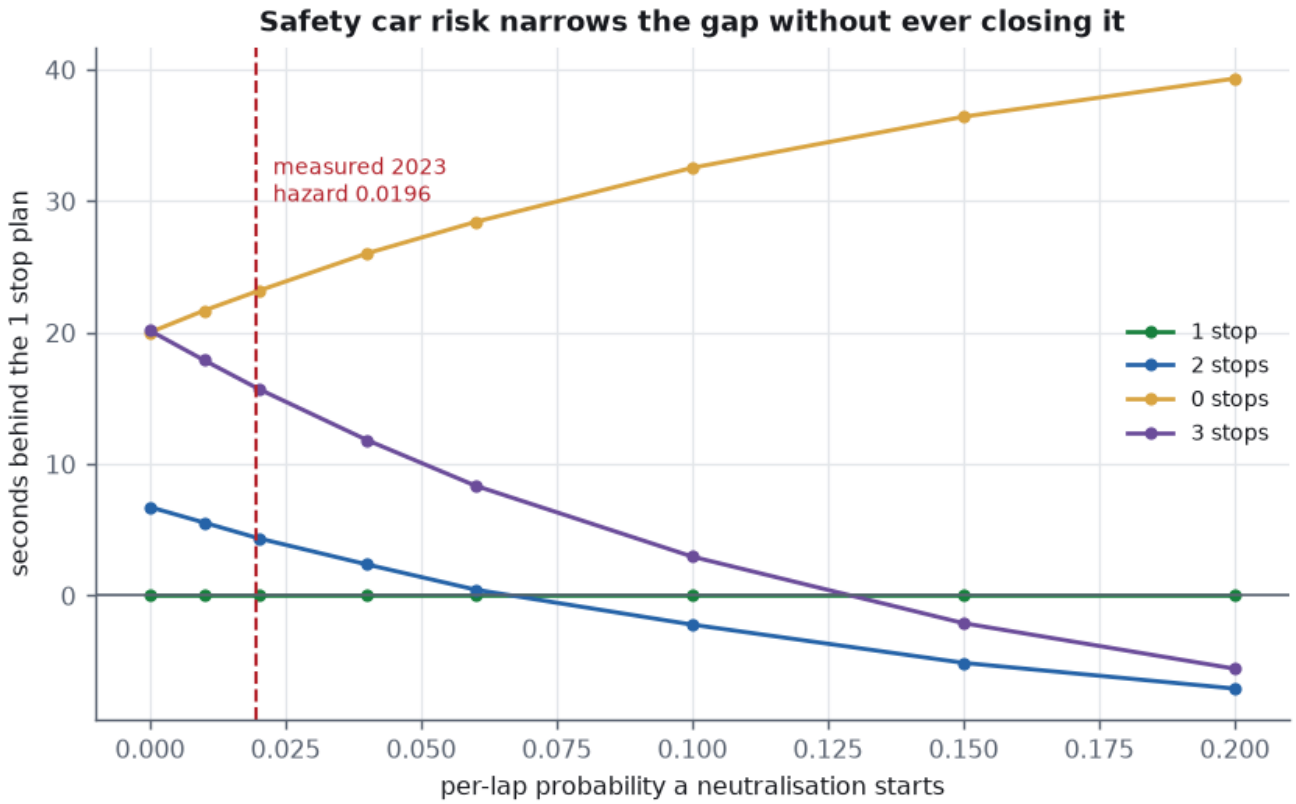




plan	stints	mean, s	sd	5th to 95th	behind best (paired, 95%)	races won
1 stop	35-35	6761.12	196.60	6570.8 to 7158.6	+0.00 ± 0.00	76.1%
2 stops	24-23-23	6765.67	194.73	6577.5 to 7158.3	+4.55 ± 0.11	22.2%
3 stops	18-18-17-17	6776.98	193.14	6591.0 to 7167.1	+15.86 ± 0.10	1.7%
0 stops	70	6784.32	199.35	6590.8 to 7189.6	+23.20 ± 0.07	0.0%

And across every hazard rate

Across hazards from zero to 0.20 — the last of which neutralises almost every race — 2 plan ever wins: 1 stop, 2 stops. Safety car risk narrows the gap between one stop and two steadily, and never closes it. So it is not the explanation for the field stopping more often.



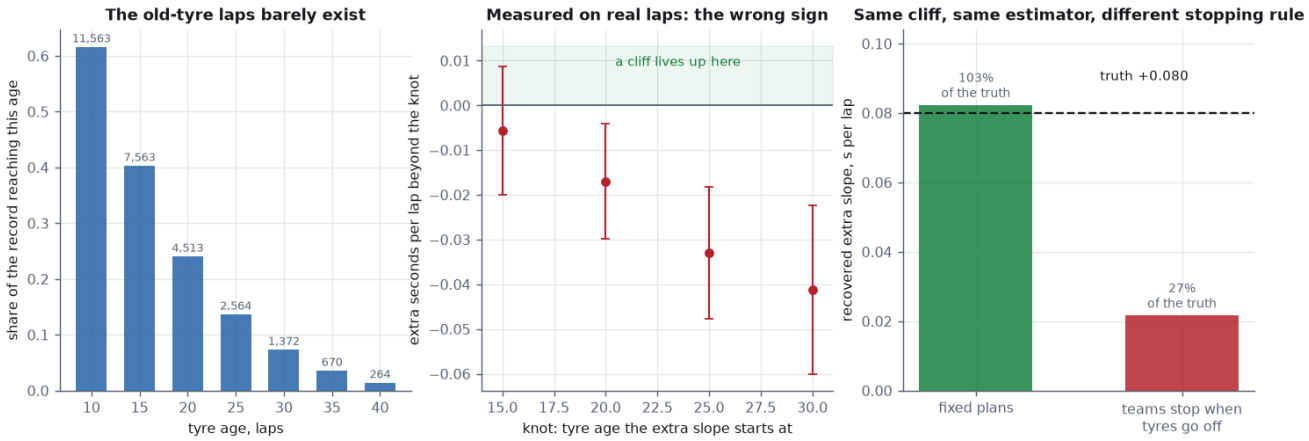
per-lap hazard	P(any neutralisation)	best plan
0.000	0.0%	1 stop
0.010	50.8%	1 stop
0.020	75.9%	1 stop
0.040	94.3%	1 stop
0.060	98.9%	1 stop
0.100	100.0%	2 stops
0.150	100.0%	2 stops
0.200	100.0%	2 stops

Reaction window, for the best plan: 0 laps → 1 stop (6762.99 s); 2 laps → 1 stop (6762.46 s); 4 laps → 1 stop (6761.92 s); 6 laps → 1 stop (6761.49 s); 8 laps → 1 stop (6761.16 s); 12 laps → 1 stop (6760.68 s)

The penalty the lap times cannot see

The remaining candidate is the one every engineer names first: tyre performance is not linear in age.

> Fitting an extra slope beyond a knot tests it directly, and the answer comes back with the wrong sign, significantly. Teams choose when to stop, and they stop before the tyre falls away, so a stint only reaches a high age when that set happened to be holding on — selection directly on the thing being measured. The laps that would reveal a cliff are the laps strategy exists to avoid.



knot	extra s/lap beyond it	95% interval	p	laps beyond
15	-0.0056	-0.0200 to +0.0087	0.441	6,857 (36.5%)
20	-0.0170	-0.0298 to -0.0041	0.010	4,062 (21.6%)
25	-0.0330	-0.0476 to -0.0183	0.000	2,274 (12.1%)
30	-0.0412	-0.0601 to -0.0223	0.000	1,198 (6.4%)

Laps reaching each tyre age: age 10: 11,563 (61.61%); age 15: 7,563 (40.30%); age 20: 4,513 (24.04%); age 25: 2,564 (13.66%); age 30: 1,372 (7.31%); age 35: 670 (3.57%); age 40: 264 (1.41%)

The same argument where the truth is known

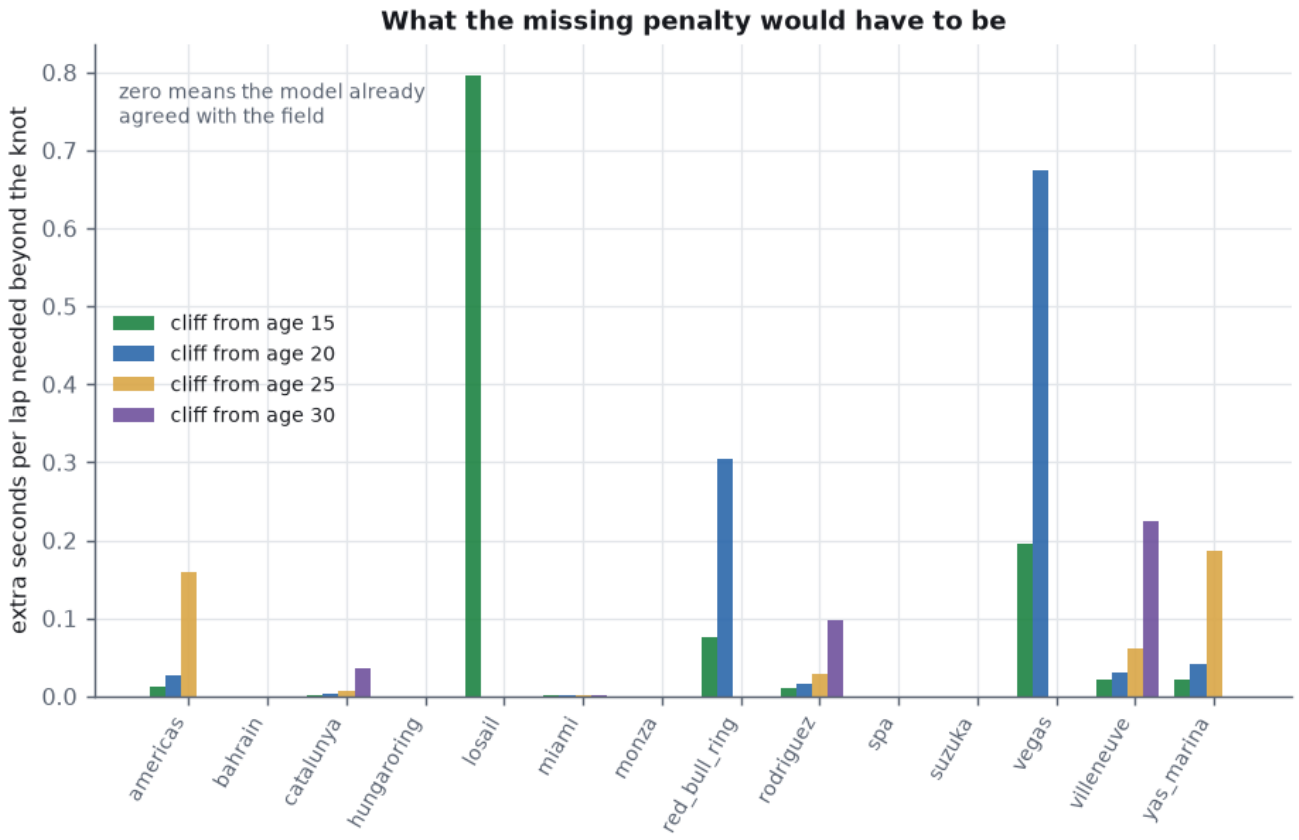
Every car is given the same real cliff of +0.0800 s per lap beyond age 20, and the two arms differ only in how stints end. With fixed plans the estimator recovers the cliff almost exactly. With a stopping rule that retires the worst sets first it recovers a fraction of it — and the more promptly teams are assumed to react, the less survives.

stints ended by	true extra slope	recovered	95% interval	share of truth	longest stint
fixed plans	+0.0800	+0.0823	+0.0777 to +0.0868	103%	30
teams stop when tyres go off	+0.0800	+0.0216	+0.0157 to +0.0276	27%	40

By how promptly teams react: stop at 1.0 s/lap → +0.0131 (16% of truth); stop at 1.3 s/lap → +0.0191 (24% of truth); stop at 1.6 s/lap → +0.0216 (27% of truth); stop at 2.0 s/lap → +0.0244 (30% of truth); stop at 2.5 s/lap → +0.0327 (41% of truth); stop at 3.0 s/lap → +0.0427 (53% of truth)

How big would it have to be?

Since the cliff cannot be measured from these laps, the question is inverted: what extra penalty would make the field's own choice the optimal one? This is not a measurement of the cliff and is not offered as one. It is a statement of the form: if a cliff is the only thing missing, it is this big.



circuit	field ran	from age 15	from age 20	from age 25	from age 30
americas	2	+0.012	+0.026	+0.158	no cliff reaches
bahrain	2	already agrees	already agrees	already agrees	already agrees
baku	1	refused	refused	refused	refused
catalunya	2	+0.002	+0.003	+0.006	+0.037
hungaroring	2	already agrees	already agrees	already agrees	already agrees
interlagos	4	refused	refused	refused	refused
jeddah	1	refused	refused	refused	refused
losail	3	+0.795	no cliff reaches	no cliff reaches	no cliff reaches
miami	1	+0.001	+0.001	+0.001	+0.002
monza	1	already agrees	already agrees	already agrees	already agrees
red_bull_ring	3	+0.075	+0.304	no cliff reaches	no cliff reaches
rodriguez	2	+0.011	+0.015	+0.029	+0.098
silverstone	1	refused	refused	refused	refused

How big would it have to be? (continued)

circuit	field ran	from age 15	from age 20	from age 25	from age 30
spa	2	already agrees	already agrees	already agrees	already agrees
suzuka	2	already agrees	already agrees	already agrees	already agrees
vegas	2	+0.196	+0.674	no cliff reaches	no cliff reaches
villeneuve	2	+0.022	+0.031	+0.061	+0.225
yas_marina	2	+0.021	+0.041	+0.186	no cliff reaches

Lap times and pit stops from the Jolpica mirror of the Ergast schema. The cleaning rules and the fixed-effects estimator are carried over from the stint-degradation project.